

Temporal Fusion Transformer for Solar Power Forecasting in Spain

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Abstract

As solar power supply to the Spanish electricity grid rises among rapid and increasing investment in the renewable energy sector, the need for timely and informative power forecasts rises with it. In this paper, we implement a Temporal Fusion Transformer (TFT) model and compare it to, and later combine it with, XGBoost to investigate the effect of both model family and inputs type on forecasting performance. We record consistent improvement in R^2 and $nRMSE$ across all study regions when including Numerical Weather Prediction (NWP) data in the input features, but we find no evidence of a strong influence of model architecture on forecasting error. Despite this, the TFT's attention mechanisms allow for interpretability of variable importance, which may prove useful to grid operators in a production setting.

Keywords: Solar power forecasting, Temporal Fusion Transformer, XGBoost, Numerical weather prediction, Deep Learning, Day-ahead forecasting, Multi-horizon time series

1. INTRODUCTION

Solar photovoltaic (PV) capacity has seen the largest absolute growth of any renewable energy source globally, with installed capacity on track to surpass coal by 2027 [19]. Spain has been at the forefront of this expansion, recording a 17% increase in solar PV generation in recent years—one of the largest national increases worldwide [22]. With more than 6,000 photovoltaic plants distributed across 17 autonomous communities, provincial-scale generation forecasting is operationally relevant for transmission system operators (TSOs) managing grid balance and wholesale market participation. Forecast errors translate directly into balancing costs; improving accuracy by even a few percentage points has measurable economic value at grid scale [9].

The day-ahead forecasting task is challenging for two distinct reasons. First, solar generation depends on irradiance, which is only partially predictable from numerical weather prediction (NWP) models at the 24-hour horizon. Second, even with accurate weather inputs, the model must learn a non-linear mapping from meteorological fields to aggregated power output across spatially distributed plant portfolios. These two sources of error—meteorological forecast uncertainty and model approximation error—are typically conflated in existing evaluations, since most studies either fix the model family and vary the inputs, or fix the inputs and vary the model.

We design a controlled study to disentangle these effects. For each of three Spanish provinces representing climatically distinct solar regimes, we train six model variants that cross two architectures—a Temporal Fusion Transformer (TFT) [14] and gradient-boosted regression trees [7]—with three feature regimes: a *Standard* regime using only deterministically computable calendar and solar geometry features, an *NWP* regime that adds short-range numerical weather prediction forecasts as known-future inputs, and a *Perfect Forecast* regime that substitutes ERA5-Land reanalysis data as an empirical performance upper bound. The TFT is particularly suited to this comparison: its architecture explicitly distinguishes between observed past inputs and known future inputs, and its variable selection networks provide interpretable importance weights that reveal how the model uses NWP covariates [16]. This design allows NWP value and architecture value to be measured independently across geographically heterogeneous test cases.

We address the following research question: *To what extent does integrating NWP forecasts as known-future covariates reduce day-ahead solar PV generation forecasting error, and does a Temporal Fusion Transformer's attention-based covariate selection provide additional value over gradient-boosted trees when both models share identical feature*

sets?

The main contributions of this work are: (i) a controlled feature-regime ablation isolating the marginal value of NWP covariates from model architecture effects; (ii) an empirical comparison of TFT and XGBoost under matched inputs across three climatically distinct provinces; and (iii) an analysis of the Perfect Forecast upper bound that reveals a covariate distribution shift effect, with practical implications for evaluating models trained on NWP forecast data.

The remainder of the paper is organised as follows. Section 2 reviews related work. Section 3 describes the data and formal problem setup. Section 4 details the methods. Section 5 describes the experimental setup. Section 6 presents and analyses results. Section 7 discusses findings and limitations.

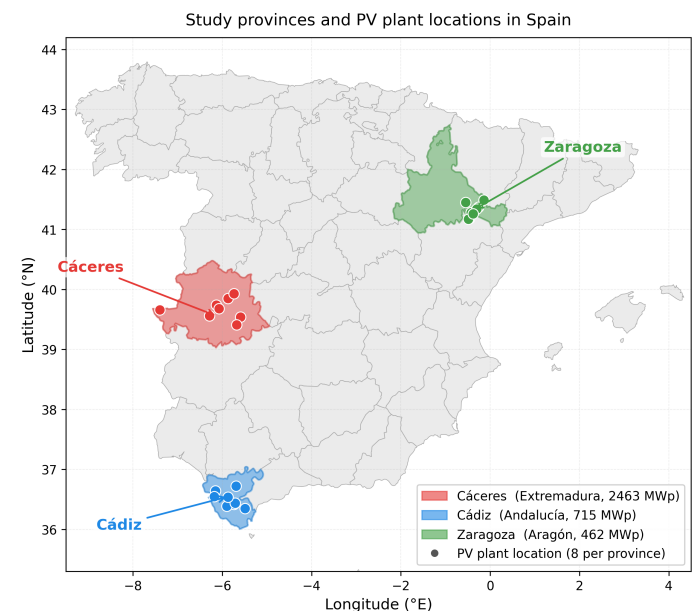


Figure 1. Study Provinces and Plants Locations

2. RELATED WORK

2.1. Temporal Deep Learning for Energy Forecasting

Recurrent architectures—particularly LSTMs [3]—have been the dominant sequence model for short-term load and generation forecasting. The Temporal Fusion Transformer [14] extended this line by combin-

ing LSTM-based local temporal processing with scaled dot-product attention [8], gated variable selection networks, and explicit handling of static metadata and known-future inputs. These design choices make TFT well-suited to forecasting tasks where future covariates of varying quality (calendar features, NWP forecasts) are available alongside historical observations, and where interpretability via attention weights and variable importance is desirable.

2.2. Gradient-Boosted Trees as Competitive Baselines

Gradient-boosted regression trees, and XGBoost in particular [7], remain strong baselines for tabular regression tasks. Grinsztajn et al. [15] demonstrated that tree-based models systematically outperform deep learning on heterogeneous tabular benchmarks, attributing the gap to the rotational non-invariance of tree splits and their robustness to uninformative features. Energy forecasting often involves a mix of smooth (calendar) and irregular (meteorological) predictors, making this comparison non-trivial. We therefore use XGBoost not as a strawman but as a strong baseline, and evaluate it under the same feature sets as TFT to separate the architecture effect from the feature effect.

2.3. Solar Forecasting and NWP Integration

Solar generation forecasting from meteorological inputs has been studied extensively [9]. Lorenz et al. [4] demonstrated that NWP-derived irradiance forecasts provide meaningful skill at the day-ahead horizon, and that spatial aggregation over plant portfolios reduces prediction error relative to single-site forecasts — both findings that inform our provincial aggregation design. Pedro and Coimbra [5] established benchmarks for solar power prediction without exogenous inputs, providing context for the baseline performance level of our Standard feature regime. More recently, Schubnel et al. [21] proposed SolarCrossFormer, a transformer model that fuses satellite imagery with ground-based sensor data for day-ahead solar irradiance forecasting, achieving 6.1% nMAE across Switzerland and demonstrating that multi-source data fusion substantially outperforms single-source models at the 24-hour horizon - a finding directionally consistent with our NWP integration results. Our work extends this prior literature by treating NWP integration as an explicit ablation variable rather than a fixed modelling choice, and by evaluating across multiple climate zones within a single national grid.

3. DATA AND PROBLEM FORMULATION

3.1. Problem Formulation

Let $y_t^{(p)} \in \mathbb{R}_{\geq 0}$ denote the aggregate solar PV generation (MWh) at province p during hour t . The forecasting task is to predict the next 24 hourly values given a 168-hour observation window:

$$\hat{y}_{t+1:t+24}^{(p)} = f(\mathbf{x}_{t-167:t}^{\text{obs}}, \mathbf{x}_{t+1:t+24}^{\text{future}}, \mathbf{s}^{(p)}) \quad (1)$$

where $\mathbf{x}^{\text{obs}}$ contains covariates observed only up to time t (measured weather), $\mathbf{x}^{\text{future}}$ contains covariates known at forecast time for the entire prediction horizon (calendar features, solar geometry, and optionally NWP forecasts), and $\mathbf{s}^{(p)}$ is a static province-level context vector. The 168-hour encoder window was chosen to capture weekly periodicity in solar generation profiles.

We study three feature regimes, differing only in the content of $\mathbf{x}^{\text{future}}$: a *Standard* regime using only deterministically computable features (calendar and solar geometry), an *NWP* regime that adds short-range numerical weather prediction forecasts, and a *Perfect Forecast* regime that substitutes ERA5-Land reanalysis data in place of NWP forecasts. The Perfect Forecast regime is not operationally deployable—reanalysis is produced post-hoc—but serves as an empirical upper bound on what forecast weather inputs can achieve.

3.2. Data Sources

Generation data were obtained from the Spanish electric grid operator (ESIOS/REE) at hourly resolution for the period January 2023 to December 2025 [20]. For each province, we selected the eight plants with the highest installed capacity and collected hourly ERA5-Land reanalysis fields at each plant’s coordinates: 2 m temperature (T_{2m}), 2 m dewpoint temperature (T_d), surface pressure, total precipitation, surface solar radiation downwards (SSRD), and surface thermal radiation downwards (STRD). ERA5-Land is produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) and distributed via the Copernicus Climate Data Store [13, 18]; it provides land-surface variables at 0.1 degrees spatial resolution by replaying the land component of ERA5 with enhanced detail. NWP forecast fields covering the same meteorological variables were obtained from Open-Meteo [17] and correspond to forecasts issued at 00:00 UTC for each forecast day, giving a forecast horizon that matches the 24-hour prediction window.

Solar geometry (zenith angle, azimuth angle) and clear-sky irradiance components (GHI, DNI, DHI) were computed from plant coordinates and timestamps using a clear-sky irradiance model [10]. The clearsky ratio $k_t = \text{SSRD}/\text{GHI}_{\text{clear}}$ was included as a cloudiness proxy; dewpoint depression $\Delta T = T_{2m} - T_d$ was included as a humidity indicator.

Table 1 summarizes all data sources.

Table 1. Summary of data sources.

Source	Variables	Resol.	Period
ESIOS	Solar PV generation	Hourly	2023–2025
ERA5-Land	T_{2m} , T_d , pressure, precip., SSRD, STRD	Hourly	2023–2025
Open-Meteo	T_{2m} , T_d , pressure, precip., SSRD, STRD (forecast)	Hourly	2023–2025
pvlb	Zenith, azimuth, $\text{GHI}_{\text{clear}}$, $\text{DNI}_{\text{clear}}$, $\text{DHI}_{\text{clear}}$	Hourly	Computed

The three study provinces span Spain geographically and are characterized by distinct climate regimes: Cáceres (Extremadura, inland semi-arid, 2,463 MWp), Cádiz (Andalucía, Atlantic coastal, 715 MWp), and Zaragoza (Aragón, Ebro valley continental, 462 MWp).

Table 2. ERA5-Land climate statistics for the three study provinces (2023–2025). All-hour means except SSRD and k_t , which are daytime-only (zenith $< 90^\circ$).

Statistic	Cáceres	Cádiz	Zaragoza
Mean temperature ($^\circ\text{C}$)	17.3	19.2	15.5
Temperature range: min–max ($^\circ\text{C}$)	−0.2–41.1	2.8–40.4	−2.8–40.6
Dew-point depression ($^\circ\text{C}$)	8.5	6.8	8.7
Annual precipitation (mm yr^{-1})	716	561	397
Mean surface pressure (hPa)	973	1009	954
Mean daytime SSRD (W m^{-2})	395	431	389
Mean clearsky index k_t	0.848	0.866	0.892
Overcast fraction ($k_t < 0.5$, %)	18.4	15.5	12.6

3.3. Spatial Aggregation

ERA5-Land fields were extracted at each plant’s coordinates and aggregated to a single province-level signal using capacity-weighted averaging:

$$x_t^{(p)} = \frac{\sum_{i=1}^8 c_i x_{i,t}}{\sum_{i=1}^8 c_i} \quad (2)$$

where c_i is the installed capacity of plant i . This weights larger plants more heavily, reflecting their proportionally larger contribution to re-

gional generation, following standard practice in regional solar power aggregation [4].

3.4. Temporal Splits and Normalization

Data were split chronologically into three non-overlapping partitions: training (two full years, 2023-2024; 17,544 hours), validation (2025-01-01 to 2025-06-30; 4,344 hours), and test (2025-07-01 to 2025-12-31; 4,416 h). All temporal ordering was preserved; no future information leaks into earlier partitions. Z-score normalization parameters (mean and standard deviation per feature) were computed on the training partition only and applied identically to validation and test data.

3.5. Feature Regimes

Calendar features were encoded cyclically: for a period- K quantity k , we compute $\sin(2\pi k/K)$ and $\cos(2\pi k/K)$, yielding continuous representations that respect periodicity. The three feature regimes are summarized in Table 3.

Table 3. Feature regimes. Checkmarks indicate inclusion.

Feature group	Std.	NWP	Perfect
Cyclical calendar (6)	✓	✓	✓
Solar geometry (2)	✓	✓	✓
Clearsky GHI (1)	✓	✓	✓
ERA5 observed (8)	✓	✓	—
NWP forecast (7)		✓	
ERA5 as known future (8)			✓

In the Perfect Forecast regime, all eight ERA5 weather variables are reclassified from observed to known future; the observed set is empty.

4. METHODS

4.1. Temporal Fusion Transformer

The Temporal Fusion Transformer (TFT) [14] is a multi-horizon forecasting architecture that explicitly distinguishes between observed past inputs, known future inputs, and static metadata. It uses three mechanisms in sequence: variable selection networks that gate the contribution of each input feature, an LSTM encoder-decoder that captures local temporal dynamics, and a multi-head attention layer that models long-range dependencies across the encoder sequence.

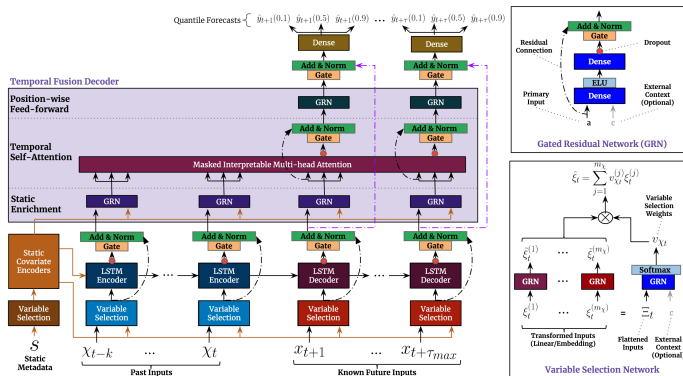


Figure 2. Temporal Fusion Transformer Architecture. Figure reproduced from Lim et al. [14].

Variable selection. Each input variable passes through a Gated Residual Network (GRN), which produces a soft selection weight. The GRN is defined as:

$$\text{GRN}(a, c) = \text{LayerNorm}(a + \text{GLU}(\eta_1)) \quad (3)$$

$$\eta_1 = W_1 \eta_2 + b_1, \quad \eta_2 = \text{ELU}(W_2 a + W_3 c + b_2) \quad (4)$$

where a is the primary input, c is an optional context vector (e.g., static metadata), GLU is a gated linear unit, and ELU is the exponential linear unit activation. This allows the model to suppress irrelevant features at each timestep adaptively.

Local temporal processing. The selected past features are processed by an LSTM encoder, and the selected future features by an LSTM decoder. The LSTM cell update equations are standard [3]:

$$\mathbf{f}_t = \sigma(W_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + b_f), \quad \mathbf{i}_t = \sigma(W_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + b_i) \quad (5)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tanh(W_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + b_c), \quad \mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \quad (6)$$

The encoder output is concatenated with the decoder sequence to form a joint context that is passed to the attention layer.

Long-range dependencies. Scaled dot-product multi-head attention [8] is applied over the full encoder-decoder sequence:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \quad (7)$$

The attention scores over the encoder provide interpretable weights indicating which historical timesteps are most informative for each forecast step.

Output and loss. The decoder output at each forecast step is projected to $Q = 7$ quantiles $\{0.02, 0.10, 0.25, 0.50, 0.75, 0.90, 0.98\}$ via a shared linear head. Training minimizes the sum of quantile (pinball) losses:

$$\mathcal{L}_q(\hat{y}, y) = \max(q(y - \hat{y}), (q - 1)(y - \hat{y})) \quad (8)$$

$$\mathcal{L} = \frac{1}{Q \cdot H} \sum_q \sum_{h=1}^H \mathcal{L}_q(\hat{y}_{t+h}, y_{t+h}) \quad (9)$$

where $H = 24$ is the prediction horizon. At evaluation time, the median ($q = 0.50$) prediction is used as the point forecast.

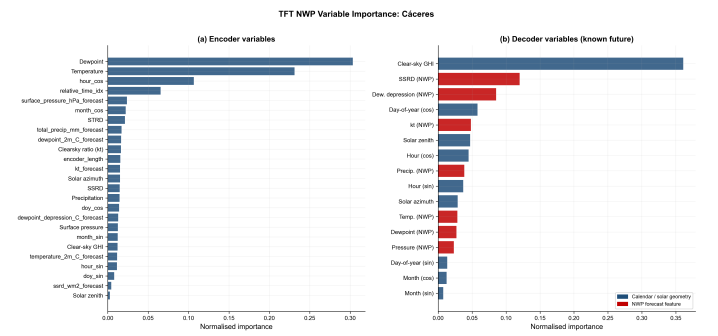


Figure 3. TFT variable importance weights for encoder (left) and decoder (right) in Cáceres under the NWP regime.

4.2. XGBoost Baseline

As a baseline, we use gradient-boosted regression trees [7] from the XGBoost (eXtreme Gradient Boosting) library. The model is an ensemble of T additive regression trees:

$$\hat{y}_t = \sum_{t=1}^T f_t(\mathbf{x}_t), \quad f_t \in \mathcal{F} \quad (10)$$

where $\mathcal{F}$ is the space of regression trees. Each tree is fit to the negative gradient of the loss from the previous iteration. The regularized training objective is:

$$\mathcal{L} = \sum_{i=1}^N \ell(y_i, \hat{y}_i) + \sum_{t=1}^T \Omega(f_t), \quad \Omega(f) = \gamma|T| + \frac{1}{2}\lambda\|w\|^2 \quad (11)$$

where $|T|$ is the number of leaves, w are leaf weights, and γ, λ are regularization hyperparameters.

Unlike TFT, XGBoost operates on fixed-length feature vectors and does not model sequence dynamics. For the multi-step forecasting task, each prediction at horizon $h \in \{1, \dots, 24\}$ is made by a separate model instance that receives a flat feature vector composed of the known-future features at timestep $t + h$ (calendar, solar geometry, and optionally NWP forecasts). No autoregressive chaining is used; each horizon is treated as an independent regression problem. This is computationally simpler but does not allow the model to condition on intermediate predictions.

4.3. Stacking Ensemble

We combine the two NWP-informed models using stacked generalization [2]. TFT and XGBoost exhibit partially uncorrelated residuals on the validation set: TFT captures long-range temporal dependencies through attention while XGBoost exploits non-linear feature interactions via tree splits, and a linear meta-learner can in principle reduce variance by weighting their complementary strengths. The primary engineering challenge was obtaining out-of-sample first-level predictions for meta-learner training without leaking test information; we used the held-out validation split (January–June 2025) for this purpose, as discussed in Section 6.5. At the first level, TFT NWP and XGBoost NWP each produce predictions $\hat{y}^{\text{TFT}}$ and $\hat{y}^{\text{XGB}}$ for the validation period. At the second level, a Ridge regression meta-learner is fitted on these predictions:

$$\hat{y}^{\text{stack}} = \beta_0 + \beta_1 \hat{y}^{\text{TFT}} + \beta_2 \hat{y}^{\text{XGB}} \quad (12)$$

The L_2 penalty in Ridge regularizes the meta-learner against overfitting to the validation period. At test time, both base models generate predictions, which are combined using the fitted coefficients.

The ensemble results are reported in Section 6.5.

4.4. Hyperparameter Optimization

Hyperparameters for all models were tuned using the Tree-structured Parzen Estimator (TPE) algorithm [11], which builds a probabilistic model of the objective surface and concentrates samples on promising regions. TPE was preferred over random or grid search because it achieves better hyperparameter-space coverage within a fixed trial budget; with 15 trials per model instance and 18 instances in total, exhaustive grid search would have been computationally prohibitive. For TFT, the objective is validation quantile loss; for XGBoost, validation mean squared error. Each model received 15 optimization trials. The test set was held out entirely during this process and evaluated only once, after the best hyperparameter configuration was selected from validation performance.

Table 4 lists the search spaces.

Table 4. Hyperparameter search spaces.

Hyperparameter	Range	Type
TFT		
hidden_size	{16, 32, 64, 128}	Categorical
lstm_layers	{1, 2}	Categorical
attention_head_size	{1, 2, 4}	Categorical
dropout	[0.1, 0.4]	Float
hid-	{8, 16, 32}	Categorical
den_continuous_size		
learning_rate	$[10^{-4}, 10^{-2}]$	Log-float
XGBoost		
n_estimators	[100, 1000]	Integer
max_depth	[3, 10]	Integer
learning_rate	[0.01, 0.3]	Log-float
subsample	[0.6, 1.0]	Float
colsample_bytree	[0.6, 1.0]	Float
min_child_weight	[1, 10]	Integer
reg_alpha /	$[10^{-8}, 10]$	Log-float
reg_lambda		

5. EXPERIMENTAL SETUP

5.1. Evaluation Metrics

We report two metrics. The coefficient of determination:

$$R^2 = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \bar{y})^2} \quad (13)$$

measures the proportion of variance in the target explained by the model, with $R^2 = 1$ indicating perfect prediction. The normalized root mean squared error:

$$\text{nRMSE} = \frac{\sqrt{\frac{1}{N} \sum_{t=1}^N (y_t - \hat{y}_t)^2}}{\bar{y}_{\text{train}}} \times 100\% \quad (14)$$

normalizes the RMSE by the mean training generation, making it comparable across provinces with different installed capacities and usage patterns. Absolute RMSE in MWh would conflate model skill with regional scale, so nRMSE is the primary metric for cross-province comparisons.

To quantify sampling uncertainty in test-set evaluation without retraining, we computed 95% bootstrap confidence intervals by resampling test-set predictions with replacement ($n = 10,000$ iterations) and recalculating R^2 and nRMSE on each resample. These intervals reflect test-set sampling variance for fixed trained models; they do not capture variance from model initialization or hyperparameter selection [1]. For the bar chart in Figure 4, we plot ± 1 bootstrap standard deviation as error bars.

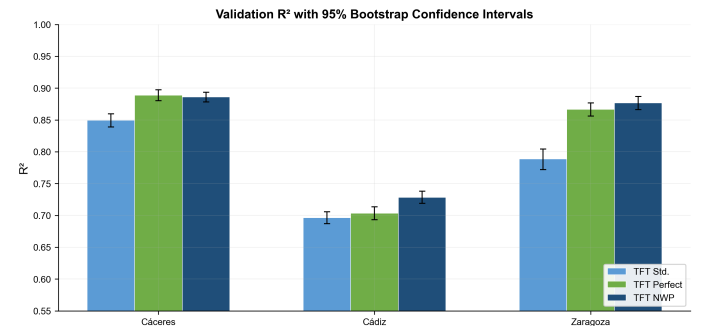


Figure 4. Test/Validation R^2 with 95% C.I. Error bars show ± 1 bootstrap standard deviation (1 000 resamples of test predictions).

5.2. Evaluation Design

The study produces 18 model instances: 6 model variants $\times$ 3 provinces. The comparison is structured to isolate two effects. To

isolate the *NWP effect*, we compare Standard and NWP variants within each architecture (e.g., TFT Standard vs. TFT NWP) – the only difference is the content of $\mathbf{x}^{\text{future}}$. To isolate the *architecture effect*, we compare TFT and XGBoost within each feature regime (e.g., TFT NWP vs. XGBoost NWP) – the only difference is the model family.

5.3. Training Configuration

TFT models were trained for up to 75 epochs with early stopping (patience 10 epochs) on validation quantile loss. Optimization used Adam [6], an adaptive gradient descent method that maintains per-parameter learning rates scaled by estimates of first and second moments of the gradients, with gradient norm clipping at 0.1 and a batch size of 64. XGBoost models were trained until validation loss converged, with early stopping after 50 rounds without improvement.

6. RESULTS

6.1. Effect of NWP Integration

Table 5 and Table 6 report test-set R^2 and nRMSE for all 18 model instances.

Table 5. Test-set R^2 for all model–province combinations.

Model	Cáceres	Cádiz	Zaragoza	Avg.
TFT Std.	0.834	0.658	0.850	0.781
TFT NWP	0.891	0.750	0.897	0.846
TFT Perfect	0.876	0.650	0.894	0.807
XGB Std.	0.801	0.715	0.813	0.776
XGB NWP	0.893	0.762	0.891	0.848
Stacking	0.784	0.935	0.907	0.875

Stacking ensemble: Ridge meta-learner trained on validation predictions; see Section 6.5 for caveats.

Bootstrap 95 % confidence intervals (10,000 resamples of test predictions) are plotted as error bars in Figure 4.

Table 6. Test-set nRMSE (%) for all model–province combinations. Normalized by province-level mean training generation (Cáceres 554.9 MWh; Cádiz 228.7 MWh; Zaragoza 335.7 MWh).

Model	Cáceres	Cádiz	Zaragoza	Avg.
TFT Std.	52.4	116.4	57.7	75.5
TFT NWP	42.6	99.5	47.8	63.3
TFT Perfect	45.4	117.9	48.6	70.6
XGB Std.	57.4	106.4	64.4	76.1
XGB NWP	42.1	97.3	49.3	62.9
Stacking	59.7	50.6	45.5	51.9

Cádiz nRMSE values exceed 100% because the normalizing denominator (mean generation over all training hours, including nighttime zeros) is small relative to the error magnitude for this lower-capacity province.

Adding NWP forecasts to the known-future feature set consistently improves performance across all three provinces and both architectures. For TFT, the R^2 gain from Standard to NWP is +0.056 in Cáceres, +0.092 in Cádiz, and +0.047 in Zaragoza. For XGBoost, the corresponding gains are +0.092, +0.047, and +0.077. In terms of nRMSE, NWP variants reduce error by approximately 10 percentage points in Cáceres and Zaragoza, and by 17 percentage points in Cádiz. The improvement is universal: no province or architecture regresses with NWP inputs.

The magnitude of the NWP benefit varies geographically and its pattern differs between architectures. For TFT, the largest absolute R^2 gain occurs in Cádiz (+0.092); for XGBoost, in Cáceres (+0.092). The larger NWP gain for TFT in Cádiz is partly a baseline effect: TFT starts from a lower Standard-regime score in this province (0.658 vs. 0.715 for XGBoost), so a larger absolute gain is required to reach comparable NWP performance. Despite this gain, TFT NWP (0.750) remains marginally below XGBoost NWP (0.762) in Cádiz, suggesting

the architecture does not confer a systematic advantage in the NWP regime for this coastal province.

6.2. Architecture Comparison Under Matched Features

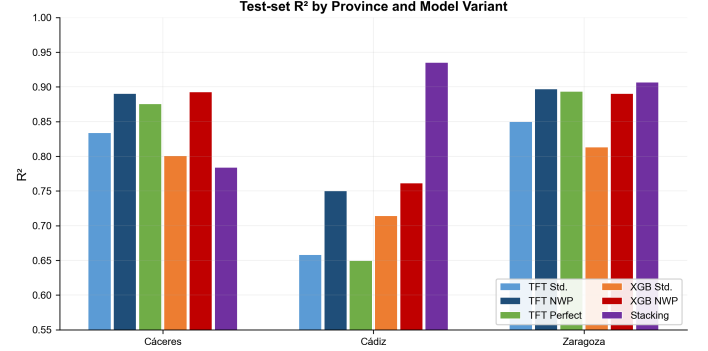


Figure 5. Test-set R^2 by province and model variant.

Within the Standard feature regime, TFT and XGBoost achieve nearly identical average R^2 (0.781 vs. 0.776). In Cádiz specifically, XGBoost Standard ($R^2 = 0.715$) outperforms TFT Standard ($R^2 = 0.658$) by a margin of 0.056. Atlantic-facing coastal regions of the Iberian Peninsula exhibit higher solar irradiance variability than interior locations due to the influence of maritime air masses [12], producing irregular meteorological patterns that a tabular model handles more robustly than a sequence-based TFT when only deterministic calendar and solar geometry features are available.

Within the NWP regime, both architectures converge. XGBoost NWP achieves a marginally higher average R^2 (0.848) than TFT NWP (0.846) – a difference of 0.002. In Cádiz, XGBoost NWP (0.762) slightly outperforms TFT NWP (0.750); in Cáceres the models are essentially tied (0.893 vs. 0.891); only in Zaragoza does TFT NWP hold a small lead (0.897 vs. 0.891). Taken together, these results suggest that the architecture – attention-based versus tree-based – contributes negligibly once NWP forecasts are included. The dominant source of improvement is the feature information itself, not the model’s capacity to exploit temporal structure. This finding is consistent with Grinsztajn et al. [15], who showed that tree-based models match deep learning on tabular regression tasks; our results extend that observation to a structured temporal setting when informative future covariates are available.

6.3. Perfect Forecast as an Upper Bound

A counterintuitive result emerges from the Perfect Forecast condition. TFT Perfect Forecast, which uses ERA5-Land reanalysis as known-future weather, achieves a lower average R^2 (0.807) than TFT NWP (0.846). The discrepancy is most pronounced in Cádiz: TFT Perfect achieves $R^2 = 0.650$, substantially below both TFT NWP (0.750) and XGBoost Standard (0.715). Notably, TFT Perfect underperforms TFT NWP in all three provinces not only in the most variable one, suggesting a systematic rather than province-specific effect.

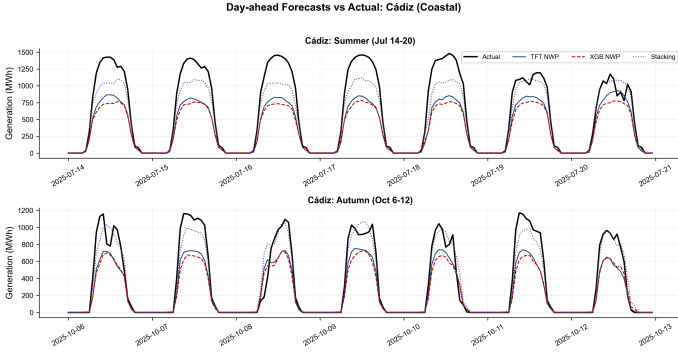


Figure 6. Hourly generation forecasts for Cádiz during two representative high-variability weeks. Actual (black), TFT NWP (blue), XGBoost NWP (red), meta-learner (purple).

This result merits careful interpretation. We identify two plausible explanations, though we cannot definitively isolate their individual contributions. First, the Perfect Forecast regime substitutes ERA5-Land reanalysis for NWP forecast fields in $\mathbf{x}^{\text{future}}$, but the TFT Standard and NWP models were trained with ERA5 weather as observed past inputs in $\mathbf{x}^{\text{obs}}$. In the Perfect Forecast regime, the same ERA5 data source therefore appears in both the encoder (as historical context) and decoder (as known-future input), which may weaken the distributional distinction between past and future that the encoder-decoder architecture is designed to exploit.

Second, Open-Meteo NWP forecasts may carry characteristic systematic biases including: temporal smoothing, location-dependent model selection, and potentially finer spatial resolution than ERA5-Land’s fixed 0.1° grid (Open-Meteo automatically selects the highest-resolution NWP model available for each location) that are structurally distinct from ERA5 reanalysis. If the TFT has learned to leverage these forecast-specific patterns as an implicit signal, substituting ERA5 would disrupt that calibration. Rather than a definitive failure of the Perfect Forecast design, this result suggests a sensitivity of the TFT to the distributional properties of its future-covariate inputs, with practical implications for pipelines that mix reanalysis and forecast data. A clean ceiling estimate would require training a separate TFT model from scratch with ERA5 fields as known-future inputs throughout, so that the variable selection networks and encoder-decoder calibrate to the distributional properties of reanalysis rather than forecast data. Without this, the comparison conflates the information content of ERA5 with a possible covariate shift penalty, and the resulting R^2 may underestimate what perfect weather knowledge could achieve in principle.

6.4. Geographic Variation

Forecast skill varies substantially across provinces. Zaragoza achieves the best R^2 across nearly all model variants (TFT NWP: 0.897; XGBoost NWP: 0.891), consistent with its continental Ebro valley climate: low humidity, low cloud fraction, and high mean clearsky index ($k_t = 0.892$, Table 2). The solar resource is highly predictable from calendar and meteorological variables alike.

Cádiz is the most challenging province, with all models achieving lower R^2 than the other two regions under the same feature regime. The Atlantic coastal location produces frequent low-cloud episodes and rapid irradiance ramps at the boundary of maritime air masses that are not fully captured by the ERA5-Land spatial resolution or the Open-Meteo NWP fields. Despite this, TFT NWP still achieves $R^2 = 0.750$ in Cádiz, a meaningful improvement over TFT Standard (0.658), further supporting the observation that NWP is informative even in the most variable regime. Cádiz nRMSE values exceed 100% in several conditions (Table 6) because the normalizing mean generation (228.7 MWh, averaged over all training hours including nighttime zeros) is small relative to the absolute error for this lower-capacity

province; this does not reflect a qualitatively different level of model failure but rather the scale sensitivity of the nRMSE denominator.

Cáceres, the largest province by installed capacity (2,463 MWp), falls between the two. Both NWP models achieve $R^2 \approx 0.891$ – 0.893 here, the strongest absolute nRMSE reduction of any province (~ 10 percentage points relative to the Standard regime).

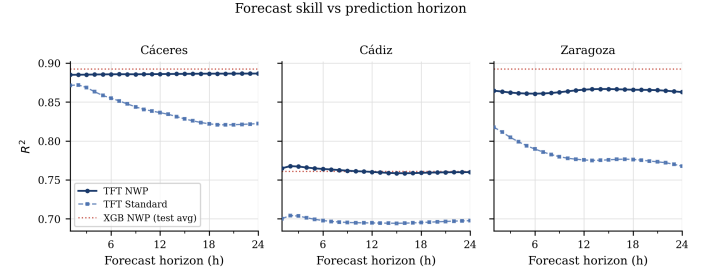


Figure 7. R^2 as a function of forecast horizon ($h = 1, \dots, 24$) for TFT NWP and XGBoost NWP across all three provinces. The XGBoost NWP line is flat because each horizon is predicted by an independent model instance with identical feature quality; unlike TFT, there is no autoregressive or sequential mechanism causing performance to degrade with increasing lead time.

6.5. Stacking Ensemble

The stacking ensemble combines the median TFT NWP predictions with XGBoost NWP predictions via a Ridge regression meta-learner fitted on out-of-sample validation predictions. Results are highly inconsistent across provinces: the ensemble achieves the highest R^2 in the study in Cádiz (0.935, vs. TFT NWP base model of 0.750), a marginal improvement in Zaragoza (0.907 vs. 0.897), and a substantial degradation in Cáceres (0.784 vs. 0.891).

These results should be interpreted with care. The meta-learner is trained on the validation period (January–June 2025) and tested on July–December 2025, introducing a seasonal distribution shift: the validation set is dominated by winter and spring solar patterns, while the test period covers peak-summer production and autumn ramp-down. Our generation records begin in January 2023, as several plants were not yet at full operational capacity prior to that date and provincial output changed substantially as new capacity came online; this constrained the training window to two years and prevented a full-year validation split that would have provided a more seasonally balanced meta-learner. The extreme stacking gain in Cádiz (+0.185 over the best base model) alongside simultaneous degradation in Cáceres (−0.107) is a signature of this overfitting: the meta-learner appears to have learned seasonal patterns specific to the validation period rather than combining complementary model residuals.

7. DISCUSSION

NWP integration is the dominant factor. Across all provinces and both architectures, providing short-range NWP forecasts as known-future covariates consistently reduces forecast error. The average nRMSE improvement from Standard to NWP variants is 12.2 percentage points for TFT and 13.2 for XGBoost. This finding aligns with prior work on NWP-informed solar forecasting [4, 5], confirming its relevance for grid-scale generation prediction. The improvement is universal: no province or architecture regresses with NWP inputs, and the benefit persists even in Cádiz, the most meteorologically variable province in the study.

Architecture matters less than expected. Within matched feature regimes, TFT and XGBoost perform nearly identically. The average R^2 difference in the NWP regime is 0.002 in favour of XGBoost (0.848 vs. 0.846), well within the range of variation attributable to hyperparameter search and random initialization. Grinsztajn et al. [15]

observed similar parity between tree-based and deep learning methods on tabular regression tasks; our results suggest this extends to structured temporal forecasting when informative future covariates are available. The TFT’s attention mechanism offers interpretability benefits through variable importance weights (Figure 3) and temporal attention profiles, but does not translate into systematic predictive gains in this setting. For practitioners, this implies that the choice between TFT and XGBoost for day-ahead solar forecasting should be driven by interpretability requirements and engineering constraints rather than expected accuracy differences.

The Perfect Forecast result and covariate distribution shift. The underperformance of TFT Perfect Forecast relative to TFT NWP is best understood as a covariate distribution shift effect, as discussed in Section 6. Both ERA5 and NWP share the same variable set (Table 3), so the effect is not about feature mismatch but about the statistical properties of the data source. This has a practical implication: researchers using reanalysis data as a proxy for “perfect forecasts” should be aware that the resulting performance may underestimate the true ceiling, particularly for encoder–decoder architectures that are sensitive to the distributional properties of their inputs. A clean performance ceiling would require a separate model trained from scratch on ERA5 future inputs, calibrated to ERA5’s distributional properties from the outset.

Geographic variation and operational implications. Forecast skill varies substantially across provinces, driven primarily by climate regime rather than installed capacity. Zaragoza (continental, $k_t = 0.892$) is the most predictable; Cádiz (Atlantic coastal, $k_t = 0.866$) is the most challenging, with nRMSE values exceeding 100% under several conditions due to the low mean generation denominator. This geographic variation suggests that NWP integration is most valuable precisely where forecasting is hardest: the largest absolute R^2 gain from Standard to NWP for TFT occurs in Cádiz (+0.092), the province with the lowest baseline performance.

Computational Limitations. The hyperparameter search budget of 15 Optuna trials is modest for a 5–6 dimensional search space; the selected configurations are likely suboptimal, and the performance gap between architectures may partly reflect differential sensitivity to random initialisation. We report single-run results due to computational constraints; TFT training variance across random seeds is known to be modest for converged models [14], but this limits statistical claims about small performance differences such as the 0.003 R^2 gap between TFT NWP and XGBoost NWP. The test set covers only July–December 2025, so winter and spring solar dynamics are absent from the holdout evaluation. No naive persistence or climatological baseline is included, limiting the interpretability of absolute nRMSE values. Finally, the stacking ensemble results are compromised by the seasonal distribution shift between validation (January–June) and test (July–December), as evidenced by the inconsistent province-level results discussed in Section 6.5.

Data limitations. The Spanish electricity system comprises over 6,000 registered photovoltaic plants, but publicly accessible generation data could only be obtained for approximately 600 of these at the time of the study. For each province we selected the eight largest plants by installed capacity, which cover a substantial share of provincial generation but not all of it. Scaling this approach nationally would leave the majority of installed capacity unrepresented, particularly smaller distributed installations whose generation characteristics may differ from the large utility-scale plants in our sample. This selection bias means our results are most directly applicable to provinces where a small number of large plants dominate aggregate output; in provinces with more fragmented plant portfolios, forecast skill may differ. Additionally, due to the sharp nation-wide increase in installed PV capacity, we faced several challenges in gathering consistent data

for prolonged time spans. The generation records begin in January 2023 because several plants that we eventually included in the study were not yet at full operational capacity prior to that date, constraining the usable dataset to three years and preventing a seasonally balanced validation split, a limitation that directly contributed to the stacking ensemble’s inconsistent results (Section 6.5). The limited training window is likely to disproportionately affect the TFT, whose attention mechanism, LSTM encoder, and gating layers present substantially more parameters to estimate than XGBoost’s ensemble of shallow trees; with only three years of hourly data the model may lack sufficient examples to learn the long-range temporal dependencies its architecture is designed to capture, and the near-parity we observe between TFT and XGBoost may partly reflect a data regime in which the TFT’s capacity for temporal modelling is underutilised.

Future work. Increasing the HPO budget, evaluating a full-year test period, and comparing against a persistence baseline are natural next steps. The TFT’s quantile outputs enable probabilistic forecast evaluation (e.g., reliability diagrams and interval sharpness scores), which we leave for future analysis. Cross-province transfer learning — initialising province models from a shared pre-trained encoder, may reduce the data requirements for smaller provinces such as Cádiz. The stacking ensemble should be revisited with proper out-of-fold cross-validation or a seasonally balanced validation split before drawing conclusions about ensemble value.

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